

Screening capabilities

Revision C | 09 Apr 2025 | Dr L. Benmezroun, Study Director
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Build the screening set around the decision

- Primary potency assay: rank activity under defined conditions.
- Selectivity sweep: compare off-target inhibition at a stated concentration.
- hERG automated patch clamp and 5-isoform CYP panel: identify liabilities for follow-up.
- RED PPB: measure binding with recovery checks.
- Agree the screen gate, sample requirements and reporting limits before study start.

Plate acceptance comes first

- Define control ranges and repeat rules in the study plan.
- Inspect reference response, vehicle baseline and assay signal before releasing results.
- Record exclusions and repeat reasons on the run sheet.
- Ines Alvarado coordinates acceptance records.
- Separate assay failure from a compound-specific limitation.
- Retain qualified results with their notes.

Primary potency assay

- Concentration response in nM or μ M with assay conditions stated.
- Agree ATP conditions, concentration spacing, replication and reference compound.
- Inspect curve shape and assay window before ranking.
- Require compound identity, test-lot reference and stock concentration.
- Confirm solvent compatibility and solubility across the requested range.
- Report incomplete responses as bounded or qualified results.

Selectivity sweep

- Agree the off-target kinase list and screening concentration.
- Report inhibition in percent at that concentration; preserve the full fingerprint.
- Use follow-up concentration responses for selected findings.
- Provide compatible stock sufficient for the panel and controls.
- Retain identity across potency and sweep run sheets.
- Choose representatives for the chemistry question, not only potency.

hERG automated patch clamp

- Ravi Deshmukh leads electrophysiology interpretation.
- Assess vehicle response, cell stability and E-4031 response before release.
- Report cell-level responses, summary IC50 and tested concentration limit.
- Confirm stock concentration, solvent and requested top concentration.
- Discuss solubility constraints before test-lot dispatch.
- Keep cell-count, unstable-response and censored-result notes attached.

5-isoform CYP panel

- Screen inhibition under defined substrate and incubation conditions.
- Interpret reference-inhibitor response before ranking.
- Use 3A4 concentration response for a selected follow-up question.
- Provide identity, stock concentration and solvent composition.
- Confirm material against the agreed range and repeat allowance.
- Do not turn a single screening point into an IC50.

RED PPB

- Equilibrium dialysis with matrix and incubation conditions stated.
- Check recovery before interpreting bound and unbound fractions.
- Investigate low recovery before compound ranking.
- Agree species, matrix, test concentration and stock compatibility.
- Discuss adsorption and solubility concerns before starting.
- Keep recovery with the binding result in every handover.

Sample requirements: agree before shipment

Requirement	Sample-card information
Identity	Compound identifier and test-lot reference matching dispatch list
Material	Available amount or volume and physical form
Stock	Concentration, solvent and preparation information
Handling	Storage, stability and known solubility constraints
Scope	Assays, concentration range and repeat allowance
Traceability	Contact for discrepancies and sample reconciliation

From sample receipt to screen gate

Scope and sample card agreed > Receipt reconciled > Assay run and controls > Plate acceptance > Qualified results and ranking > Client screen gate



Receipt reconciled > Assay run and controls > Plate acceptance > Qualified results and ranking

What the client receives

Readout element	Decision value
Conditions and plate acceptance	Identifies interpretable comparisons
Compound tables	Preserves identifiers, units and qualifications
Charts where study data support them	Makes screening-set trends visible
Chemistry context where available	Connects representatives to the screening question
Biology memo	Separates screen gate from mechanistic claims
Follow-up recommendation	States the question that remains open

Start with the next screening decision

- Send the screen gate and sample-card list to Leila Benmezroun, Study Director.
- Bethany Rowcroft agrees scope and operational priorities.
- Ravi Deshmukh handles hERG design; Ines Alvarado coordinates plate acceptance.
- Henning Flantz resolves sample requirements; John Mckinney coordinates delivery.
- Confirm conditions, acceptance criteria, qualifications and handover format before dispatch.